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Article

Generative AI and English essay writing: exploring the role of ChatGPT in enhancing learners' task engagement

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Despite the growing use of artificial intelligence (AI) in L2 education, limited research has explored its impact on learners' task engagement, specifically in writing tasks. This study explored the role of using generative AI-based tools, namely ChatGPT, in influencing 60 undergraduate learners' engagement in English argumentative essay writing within an English as a Foreign Language (EFL) context over a 2-month instructional period. Using a mixed-methods experimental design, the study analyzed quantitative data from control and experimental groups alongside qualitative insights from participant interviews. Findings revealed that learners using ChatGPT demonstrated significantly higher engagement levels during English essay writing tasks. Qualitative analysis pointed to several themes, including increased motivation, reduced anxiety, enhanced interest, partnership, personalized feedback, and perceived competence. Overall, the findings suggest that ChatGPT can be an effective instructional tool for enhancing learners' engagement in writing English essays. While ChatGPT fosters engagement, the study underscores the need for balanced AI integration to support independent critical thinking. These findings offer insights for L2 educators, curriculum designers, and policymakers.

Introduction

In the last 20 years, interest in task engagement within second language acquisition (SLA) research has increased, given its critical role in fostering effective learning (Christenson, Reschly and Wylie 2012; Philp and Duchesne 2016; Derakhshan and Zare 2024; Zare and Derakhshan 2024). Task engagement refers to the state of being attentive and involved in completing tasks (Christenson, Reschly and Wylie 2012; Philp and Duchesne 2016; Zare et al. 2024). It includes various aspects, such as behavioral, cognitive, affective, social, and agentic, reflecting its multifaceted nature (Philp and Duchesne 2016). Besides this, engagement is dynamic and ecologically situated (Appleton, Christenson and Furlong 2008; Philp and Duchesne 2016), which makes it an elusive construct to define, highlighting the need for a deeper understanding of how learners engage in diverse learning settings (Hiver et al. 2021). Such complexities have also resulted in a shortage of experimental and longitudinal studies on learner engagement (Appleton, Christenson and Furlong 2008). Engagement is crucial for quality learning because it necessitates action, which is a defining characteristic of engagement (Dörnyei and Kormos 2000; Hiver et al. 2021). The importance of engagement is also evident in its positive impact on self-efficacy, motivation, L2 learning, and academic success (e.g. Dörnyei 2001; Fredricks, Blumenfeld and Paris 2004; Bicket et al. 2010Wang, Wu and Wang 2024).

In SLA settings, learner engagement is especially vital, as seen in the communicative approach's focus on interaction and communicative language use (Hiver et al. 2021). However, maintaining it in second/foreign language tasks, particularly in complex activities like writing English essays, presents significant challenges. The intricate nature of English essay writing requires high levels of linguistic proficiency and cognitive resources, making it particularly demanding for L2 learners (Kormos 2023). Thus, achieving effective engagement in these tasks may require substantial cognitive, affective, behavioral, agentic, and social involvement. This suggests a strong need for interventions aimed at enhancing and sustaining engagement during English essay writing tasks.

At the same time, studies have shown that incorporating technology into L2 learning activities increases learner engagement (e.g. Guo and Wang 2024; Wang and Xue 2024). This is mainly because technology enables educators to create dynamic and interactive learning experiences, allowing for personalized approaches to learning. Consequently, students become active participants in their education, which empowers them and gives them greater control over their learning process (e.g. Challob 2021). Recent advances in generative artificial intelligence (GenAI) have opened up new possibilities for innovative language learning tools, such as chatbots, to support language learning, particularly in writing activities, thus providing new avenues for enhancing and maintaining engagement in such tasks. Because of their capabilities in producing tailored content, they have been found effective in L2 education (Su, Lin and Lai 2023; Banihashem et al. 2024; Steiss et al. 2024). One prominent tool is ChatGPT, a GenAI model, developed by OpenAI. ChatGPT can assist learners in writing tasks by providing individualized feedback based on their queries (e.g. Barrot 2023). This aligns with the recent focus on learner-centered instruction and the recognition of individual differences in learning in SLA (for a review, see Griffiths and Soruç 2021). However, alongside these promising applications, recent discussions have raised ethical, pedagogical, and epistemological concerns. Critical perspectives question issues, such as data privacy, the potential for overreliance on technology, diminished human creativity, and the challenges inherent in integrating AI in a way that genuinely supports autonomous learning (Liang et al. 2023; Lin and Crosthwaite 2024; Mahapatra 2024; Šedlbauer et al. 2024; Yan 2023; Yoon, Miszoglad and Pierce 2023).

While there has been a growing body of literature exploring the implications of GenAI tools like ChatGPT within second/foreign language learning contexts (e.g. Banihashem et al. 2024; Guo and Wang 2024; Nguyen, Le and Nguyen 2024; Steiss 2024; Wang and Xue 2024; Derakhshan 2025), significant gaps remain in understanding their influence on student-related constructs such as task engagement. Much of the existing research tends to focus on students' linguistic outcomes, often overlooking how these tools may affect their behavioral, emotional, agentic, and cognitive involvement in learning tasks. To address this gap, our study pioneers the use of ChatGPT 4.0 within an experimental, sequential explanatory design to investigate its impact on learners' engagement in writing English argumentative essays. This approach not only integrates rigorous quantitative measurements with in-depth qualitative insights but also offers contextualized evidence that enriches existing theoretical perspectives on task engagement in technology-enhanced second language acquisition (SLA) settings. By examining how ChatGPT as a GenAI tool can foster L2 learners' task engagement, the study offers practical insights into the integration of technology to support learner-centered instruction. While the study does not propose a novel instructional design, it contributes to current discussions on how emerging technologies can be meaningfully incorporated into classroom practices to enhance learners' engagement.

Literature review

Task engagement in SLA: an overview

In L2 learning contexts, the concept of task engagement is often perceived as a particular form of engagement, as opposed to a facet of it (Svalberg 2018). Hence, a distinction is made between engagement with language and task engagement. Engagement with language involves learners actively practicing and interacting with the linguistic elements or structural aspects of the language. In contrast, task engagement involves learners learning the language through the completion of specific tasks (Platt and Brooks 2002; Zare, Derakhshan and Zhang 2024). Currently, there is no agreement on the dimensions of task engagement. However, scholars differentiate between behavioral, emotional, cognitive, social, and agentic engagement (Reschly and Christenson 2012; Derakhshan and Zare 2024).

Behavioral engagement emphasizes effort and active participation in learning tasks, which is a necessity for L2 learning to take place (Philp and Duchesne 2016; Oga-Baldwin 2019). It was initially seen as a dichotomy (Anderson 1975) but now is measured via time on task, participation, and verbal contributions (Dörnyei and Kormos 2000; ; Gettinger and Walter 2012). Emotional engagement encompasses learners' affective responses, such as enthusiasm, anxiety, interest, frustration, and enjoyment, which can profoundly influence L2 writing development (Reeve 2012; Zheng and Yu 2018). Cognitive engagement pertains to sustained attention and the use of self-regulated strategies that support learning, which are crucial for developing L2 writing proficiency (Helme and Clarke 2001; Reeve 2012; Philp and Duchesne 2016; Koltovskaia 2020). Social engagement reflects learners' readiness to interact and participate in collaborative activities (Svalberg 2009; Philp and Duchesne 2016), while agentic engagement captures proactive efforts to shape the learning environment, which are key to autonomous L2 learning efforts (Reeve and Tseng 2011; Reeve 2012).

In their model of task engagement, Egbert et al. (2021) view behavioral, emotional, cognitive, social, and agentic as signs, levels, or indicators of engagement. As engagement is dynamic (Philp and Duchesne 2016), each indicator is likely to fluctuate due to internal or external stimuli. For instance, a learner's cognitive engagement may increase as one encounters a challenging task that needs further concentration. Furthermore, given the malleable nature of engagement (Zare 2023), engagement can be influenced by specific pedagogical interventions. For example, agentic engagement could grow when learners are given more control over their tasks or are allowed to customize their learning paths. Additionally, engagement is ecologically situated (Hiver et al. 2021). That is, it is context-dependent and shaped by cultural, social, and physical environments, and by the interplay between a learner's attributes and the contextual factors. For instance, emotional engagement is influenced by the immediate emotional climate of the learning environment. The interplay between these indicators suggests that engagement evolves and adapts according to individual needs, instructional strategies, and the learning environment.

Task engagement can also be examined through the lens of self-determination theory (SDT). According to SDT, self-determination or the extent to which individuals control their own learning significantly influences their overall learning experience (Ryan and Deci 2000). SDT conceptualizes learning motivation as a continuum, ranging from amotivation—the least self-directed state—to intrinsic motivation, which embodies the highest level of self-determination. Heightened engagement, therefore, is closely tied to the presence of intrinsic motivation or self-directed behavior (Mozgalina 2015). SDT stipulates that engagement flourishes when the educational environment and tasks are structured in a way that meets three basic needs: competence, autonomy, and relatedness (Ryan and Deci 2017). Hence, learners are more likely to be engaged when they feel capable and competent in their learning activities. Learners are also more engaged when they have agency or a sense of control over what and how to learn. Learners also need to feel connected with others in their learning environment. Hence, tasks that encourage collaboration and social interaction are more likely to deepen learner engagement.

Despite the robust body of research supporting engagement's centrality in SLA, there still remain gaps that need to be explored. Specifically, further research is necessary to understand how the malleability of engagement can be harnessed through interventions to create more engaging learning tasks and environments (Appleton, Christenson and Furlong 2008). There is growing evidence that incorporating technology into second language learning can improve learner engagement (e.g. Bond 2020). However, there is a need for further research to explore the impact of GenAI models such as ChatGPT on learner task engagement. The following section will specifically discuss learners' task engagement in technology-enhanced learning environments.

Task engagement in technology-enhanced learning environments

As language education becomes increasingly mediated by advanced technologies (Egbert 2020), there is a growing need to understand how such tools shape learners' engagement with learning tasks. Unlike traditional classroom settings, technology-enhanced learning contexts introduce new affordances, such as interactivity, immediacy, adaptivity, and multimodality, that can significantly alter how learners perceive and approach learning tasks (Serrano et al. 2019). These affordances can both support and hinder task engagement (Lee 2012), depending on how well they align with learners' goals, preferences, and levels of autonomy.

In these contexts, task engagement is no longer a static measure of attention or participation, but a dynamic and situated process influenced by technological features, pedagogical design, and learner characteristics (Robillos and Bustos 2023). For instance, tools that offer personalized feedback, opportunities for self-paced learning, or interactive elements (e.g. gamification, simulations, or chatbot dialogue) can promote sustained attention and deeper cognitive processing (Yin, Yin and Xu 2023). Moreover, technology-enhanced environments often provide greater learner control and flexibility (Barak and Levenberg 2016), potentially increasing motivation and emotional investment in the task (Zare et al.2025) . However, the presence of technology does not inherently lead to higher task engagement; rather, its effectiveness depends on how meaningfully it is integrated into task-based instruction (Dunn and Kennedy 2019). Understanding these dynamics is essential, especially as emerging tools such as GenAI enter mainstream educational settings.

An increasing number of studies have examined learners' task engagement within technology-enhanced learning environments. These studies have consistently emphasized how digital technologies can enrich the learning experience by fostering real-time interaction, offering multimodal input, and providing instant feedback (Bahari 2023). Such features are associated with increased learner attention, time-on-task, and cognitive investment, all of which are key indicators of task engagement (Zare and Derakhshan 2024). Recent research has also highlighted the importance of learner agency, autonomy, and personalization in technology-enhanced environments. Tools that allow learners to make meaningful choices and receive adaptive support tend to promote deeper levels of task engagement (Egbert 2020). Despite the growing interest in technology-enhanced learning, few studies have specifically examined L2 learners' task engagement within AI and GenAI-powered classrooms. This gap highlights the need for empirical studies to explore how AI and GenAI tools shape learners' engagement with learning tasks.

Argumentative essay writing and ChatGPT

Recent decades have seen a growing emphasis on academic writing proficiency and literacy, which are essential for achieving academic and professional success (Wang et al. 2023). Writing in academic genres, particularly in a non-native language, presents unique challenges, due to the need to master both linguistic and discourse conventions (Fang 2021). Among these genres, argumentative essay writing is especially critical, as it significantly contributes to overall academic literacy and success (Hartwell and Aull 2022). Besides possessing linguistic competence, developing an argumentative essay requires critical thinking, position formulation, argument development, evidence presentation, and reasoning (McNeill and Krajcik, 2009) . Such demands may cause a lot of pressure on students, which can negatively impact their learning process (Wei, Zhang and Zhang 2020). Teachers may also face significant challenges in providing individualized feedback on students' essays, due to the complexity of such tasks and the number of students they need to support (Guo, Wang and Chu 2022). Nevertheless, providing learners with personalized feedback is crucial, as it enables them to meet their distinct developmental needs when learning to write English essays (Olsen and Hunnes 2023) .

ChatGPT, developed by OpenAI, is an innovative solution, that can provide students with personalized feedback, with its advanced natural language processing abilities, adaptability, and context-specific responses (Rudolph *et al.* 2023). Such capabilities allow ChatGPT to assist learners in all the stages of essay writing, including pre-writing, during-writing, and post-writing (Guo *et al.* 2022; Rudolph *et al.* 2023; Su, Lin and Lai 2023). Furthermore, this GenAI tool has the potential to provide personalized formative feedback that varies based on how users interact with it, reflecting their unique writing styles and error patterns (Barrot 2023).

Recent studies show that ChatGPT is effective in improving students' writing skills, particularly in writing argumentative essays (e.g. Barrot 2023; Su, Lin and Lai 2023; Guo and Wang 2024; Nguyen, Le and Nguyen 2024). Its effectiveness stems from its ability to respond to various writing prompts (OpenAI 2022) and offer tailored, flexible support that caters to the diverse needs of students in their argumentative essays—ranging from basic linguistic issues to more complex issues of presentation and structuring (Guo, Wang and Chu 2022; Lingard 2023; Rudolph et al., 2023). Additionally, such chatbots can serve as digital interlocutors and address the dialogical challenges of argumentative writing by mitigating barriers like social stress (Park *et al.* 2021) . In other words, they represent a learning partner that grows and develops alongside the writer. Hence, they could be viewed from Vygotsky's (1978) socio-cultural theory. According to this theory, effective learning is achieved through interaction with more knowledgeable others (MKOs) within the Zone of Proximal Development (ZPD). Traditionally, socio-cultural theory has centered on human interactions. However, such interactions could be expanded to include GenAI chatbots, thus conceptualizing them as a virtual MKO that enhances learning and development.

While using ChatGPT for writing assistance is convenient and effective, it may hinder one's creativity, critical thinking, and the ability to express thoughts coherently and accurately—skills that are essential for effective writing (Yanning 2017; Yan 2023Yoon, Miszoglad and Pierce 2023). Several studies have raised concerns about over-reliance on generative AI and emphasized the need for learners to develop critical awareness (e.g. Banihashem et al. 2024; Šedlbauer et al. 2024).

Altogether, while ChatGPT's capabilities generate optimism towards the prospect of incorporating it into English essay writing classes, critical areas remain underexplored, specifically its influence on learners' task engagement during the writing process. This study aims to bridge the gap by exploring the role of ChatGPT 4.0 in affecting learners' engagement in writing English argumentative essays. The study addressed the following research questions:

1) Does using ChatGPT 4.0 by L2 learners significantly affect their engagement in writing English argumentative essays in short and long terms?
2) How does using ChatGPT 4.0 by L2 learners affect their engagement in writing English argumentative essays?

Methods

Setting and design

The investigation took place in Iran, where English is adopted as a foreign language. Consequently, in this English as a Foreign Language (EFL) context, opportunities to engage with English are limited, with students typically encountering English only in English courses or during interactions with international visitors. The research was carried out at a public university with students majoring in English.

The design of the study was experimental, including both experimental and control groups, with pretest, immediate, and delayed post-tests, that were implemented within a mixed-methods design (sequential explanatory). The experimental design was developed to provide an answer to the first research question, namely to explore the role of using ChatGPT by learners in influencing their engagement in English essay writing, assessing both immediate and sustaining effects. Hence, the participants were initially evaluated for their proficiency in English, writing capabilities, and task engagement. They were then distributed into experimental and control groups through random assignment, after which they received the relevant experimental or control interventions. After the interventions, both immediate and delayed post-tests were given to the participants to explore the effects of using ChatGPT on influencing their engagement in English essay-writing tasks.

Furthermore, the sequential explanatory design of the study aimed to investigate how using ChatGPT affected learners' engagement in tasks dealing with writing English argumentative essays, as posed by the second research question. To implement this design, it is necessary to gather and examine data of both quantitative and qualitative nature, where qualitative data serve to provide a deeper understanding of the quantitative outcomes.

Participants

A total of 60 EFL students from X, all of whom were native speakers of Y, were selected from an initial pool of 168 undergraduate English learners at a public university in X. The inclusion criteria were explicitly defined to ensure sample homogeneity. These consisted of: willingness to participate, English proficiency, writing capability, and task engagement. All students provided written consent following the British Educational Research Association (BERA) protocols. Participants also had to demonstrate intermediate proficiency (B1 level on the CEFR), as measured by the Oxford Online Placement Test. Their comparable writing skills were ensured through performance on a TOEFL iBT independent writing task. They were also required to score within one standard deviation from the mean on a task engagement questionnaire (TEQ). Addressing these criteria helped us maintain homogeneity and eliminate the potential influence of such variables on their engagement after the intervention. Table 1 presents the participants' demographic profile.

As Table 1 shows, the final sample comprised 22 male (36.67%) and 38 female students (63.33%), with 10 second-year (16.7%), 22 third-year (36.7%), and 28 fourth-year students (46.6%). Participants' ages ranged from 19 to 36 years ($M = 21.75$, $SD = 2.43$).

Instruments

Oxford Online Placement Test

The participants' command of English was evaluated using the Oxford Online Placement Test. It adjusts difficulty based on the test taker's responses. It includes two parts: *Use of English*, which

Table 1. Participants' demographic profile.

Characteristic	Category	Number	%
Gender	Male	22	36.67
	Female	38	63.33
Academic year	Second year	10	16.67
	Third year	22	36.67
	Fourth year	28	46.66
Age Range	19 - 36 years		
	Mean (M): 21.75		
	Standard deviation (SD): 2.43		

looks into one's grasp of grammar and vocabulary, and *Listening*, which measures one's listening comprehension. The test places students on a scale ranging from beginner (Pre-A1) to mastery levels (C2), as defined by the CEFR. At the beginning of the study, this particular test was administered to ensure that all the participating students were at a similar English language proficiency level, specifically intermediate (B1).

TOEFL iBT independent writing tasks

To evaluate learners' English writing, we used TOEFL iBT independent writing tasks. These tasks evaluate how well one can write using academic English. In such tasks, participants are typically asked to compose an essay in response to a question or statement reflecting their own experiences or opinions, within 30 minutes. The essay should be written clearly and follow a logical structure. For the purposes of this study, participants were asked to complete three such tasks. The first was given before the experimental and control interventions to confirm that all participants had comparable English writing skills. The subsequent two tasks were administered after the interventions. These writing tasks were further used as a reference for filling out the task engagement questionnaire.

Task Engagement Questionnaire

The Task Engagement Questionnaire (TEQ), developed by Zare (2023), was used to measure how engaged the learners were with their tasks (see the Supplementary Appendix). The TEQ has 25 items and breaks down task engagement into five key domains, namely affective, behavioral, agentic, social, and cognitive engagement. Each of these domains has its own set of five questions, rated on a five-point Likert scale. Behavioral engagement looks at how actively involved students are in the task, measuring their on-task behaviors. Cognitive engagement assesses how well students focus and direct their attentional resources when doing the task. Affective engagement deals with learners' emotional responses toward the task. Agentic engagement examines the extent to which students purposefully contribute to the learning process and the task. Social engagement is concerned with how students interact with one another during the language learning task.

The TEQ included three parts. The first part gathered basic information like the students' names, ages, genders, and academic years. The second part presented the actual engagement items. The third part collected details on the students' interests and availability for follow-up interviews and requested their contact information. The reliability of the TEQ was confirmed, with strong internal consistency across the sub-scales, showing Cronbach's alpha values of 0.85, 0.78, 0.81, 0.74, and 0.84, respectively (see Zare 2023).

The questionnaire was shared with the students through Google Forms. It was first used at the beginning of the study as a pretest to verify that the students had similar levels of task engagement. It was also administered right after the interventions and again after a 1-month interval following the interventions, to spot changes in learners' engagement across the experimental and control groups.

Interviews

To gain a deeper understanding of the quantitative data, we conducted online interviews personally with learners who expressed interest in being part of the interviews. We involved voluntary participants from both the experimental and control groups, 15 learners from each group. The interviews were semi-structured. In these sessions, which were held in Persian, participants shared their experiences with ChatGPT, discussing what aspects of ChatGPT fostered or hindered their engagement in English writing tasks. The questions asked during the interviews included but were not limited to: *1) Were you (dis)engaged in writing the English essays? 2) Which specific features of ChatGPT influenced your level of engagement with the assigned tasks?* Interviewees were also given the opportunity to bring up any other relevant issues they wished to address. On average, every interview took about 40 minutes. The interviews took place one week following the assignment of the delayed post-test.

ChatGPT

ChatGPT 4.0 is one of the latest versions of the Generative Pre-trained Transformer series, developed by OpenAI. It is an AI conversational agent, used to understand and respond to natural language input. With its advanced language processing capabilities, ChatGPT is designed to produce coherent and relevant responses to the prompt provided by the user. As a writing assistant, ChatGPT 4.0 can play a multifunctional role in various stages of essay writing. Access to ChatGPT 4.0 was made possible via the website "https://start.chatgot.io/."

Procedures

Figure 1 provides a visual representation of the procedures. As Figure 1 shows, a total of 168 students volunteered for the study and participated in a pretesting phase. This phase included taking the Oxford Online Placement Test, completing a TOEFL iBT independent writing task, and filling out the TEQ. The purpose of pretesting was to minimize the potential impact of differences in participants' English proficiency, writing skills, and level of task engagement. Based on their scores, 60 students with comparable proficiency in English, writing skills, and engagement were selected. These participants were then divided into experimental and control groups using stratified random sampling to ensure a balanced representation of demographic and academic variables, such as age, gender, and academic year.

Prior to the intervention, students in the experimental group attended a training session on how to effectively use ChatGPT 4.0. During this session, the instructor provided guidelines on how to use ChatGPT for idea development, editing, and proofreading. The training included detailed demonstrations, hands-on practice, and clear instructions for each stage of the process. This preparatory phase ensured that participants understood the tool's capabilities, limitations, and the specific procedures to follow.

Next, the experimental intervention was administered to the experimental group, while the control intervention was administered to the control group. Both interventions comprised 12 sessions of instruction, occurring twice weekly and taught by a single English teacher. Both interventions were centered around the topic of how to write an English argumentative essay. The instruction was structured around the 'three Ps' methodology, which included presentation, practice, and production phases. Initially, the instructor laid out the framework for developing an English argumentative essay, detailing the organization and steps of writing such essays. During the practice phase, students analyzed and practiced with models of such essays, along with controlled exercises dealing with writing English argumentative essays. Finally, in the production phase, students practiced writing English argumentative essays.

The production phase was conducted differently for the experimental and control groups. For control group learners, this phase involved working with peers in the three steps of idea development, editing, and proofreading, whereas it required working with ChatGPT 4.0 for learners in the

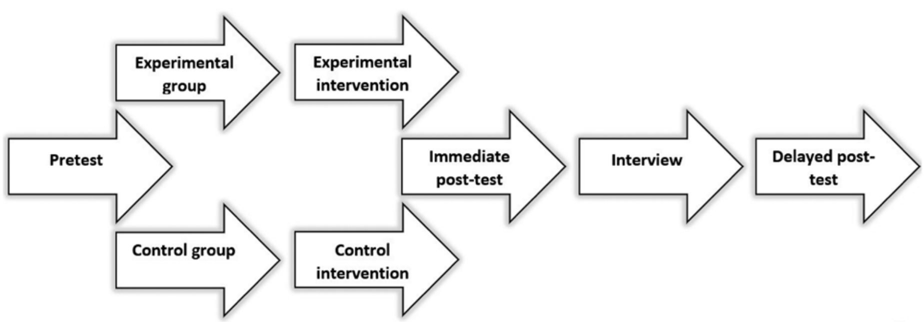

Figure 1. Procedures of the study.

experimental group. During the idea development stage, learners used ChatGPT 4.0 to develop ideas and outlines and refine them. For the editing stage, they used ChatGPT 4.0 primarily to receive feedback on their drafts. During the proofreading stage, they used ChatGPT 4.0 to detect and suggest corrections for errors in their use of words, structures, and textual devices. During the idea development, editing, and proofreading stages, peers assisted the control group learners in the same way.

To ensure that ChatGPT was used consistently and as intended, the instructor observed the students' interactions with ChatGPT and asked them to complete brief self-reports after each session. This approach helped us verify that the tool was utilized according to the study's protocol.

After the experimental and control interventions, learners in both the experimental and control groups were asked to take the TEQ twice as immediate and delayed post-testing phases. The first administration was conducted straight after the interventions as an immediate post-intervention test, while the second was conducted a month later as a delayed post-intervention test. In addition to each assignment of the TEQ, they were provided with a TOEFL writing task to refer to while answering the items in TEQ. Following the first administration of the TEQ, online interviews were held in Persian with participants from both experimental and control groups.

Data analysis

Exploring the impact of using ChatGPT 4.0 by L2 learners on their task engagement in English essay writing required an experimental sequential explanatory mixed-methods design, incorporating both quantitative and qualitative data analyses. Analysis of the quantitative data was done running parametric tests, as the pretest, immediate, and delayed posttest results verified normal distribution ($P = .162$, $.071$, $.195 > .05$). Consequently, we conducted three distinct independent-samples t-tests, one for each evaluation stage, utilizing SPSS software (version 26). Moreover, to identify any notable changes across the study's timeline—from pre-intervention to immediate and delayed post-intervention—we ran a pair of repeated-measures ANOVAs for both the control and experimental groups.

Analysis of the qualitative data required coding of the students' responses to the interviews, which were transcribed, translated into English, and checked for errors and inconsistencies. The qualitative analysis was done through MAXQDA (2022 version). To minimize preconceptions, we adopted bottom-up or emergent coding. That is, instead of analyzing the interviews with a predetermined list of codes, we derived the codes and themes from a meticulous and iterative bottom-up coding process. This involved using Braun and Clarke's (2006) model of data analysis. The model comprised five steps. First, the data were repeatedly read to gain a comprehensive understanding of their breadth and depth. Second, initial codes were created. Third, codes were categorized into broader themes. Fourth, the themes were refined, defined, and named. Finally, a report with anonymous excerpts was prepared.

To ensure the trustworthiness of the analysis, we took several measures into account. These include transferability, credibility, confirmability, and dependability (Lincoln and Guba 1985). To improve the credibility of our findings, we employed data triangulation by collecting and analyzing both quantitative and qualitative data. The quantitative data were derived from the TEQ, which provided measurable insights into learner engagement, while the qualitative data were obtained through in-depth interviews that explored participants' experiences and perceptions in greater detail. The transferability of the analysis was ensured by providing a detailed report that included information on the study's design, setting, subjects, instruments, and themes. Dependability was enhanced through member checking or participant validation (Creswell 2014). Specifically, five participants were asked to check if the derived themes reflected their opinions. To address confirmability, two coders coded the data independently, and their inter-coder reliability was computed, using Cronbach's alpha ($\alpha = 0.90$). Another coder from a foreign context was also recruited to code and check a sample of the data ($\alpha = 0.88$), aiming to enhance the confirmability of the analysis. Inviting the foreign coder helped us maintain both emic and etic perspectives, thus addressing researcher positioning.

Ethical considerations

Concerning ethics, ensuring informed consent and maintaining participant confidentiality was paramount at every stage of our analysis. We asked for participants' consent prior to data collection. Moreover, we preserved their anonymity and privacy during analysis and write-up. The names in the excerpts are pseudonyms.

Results

Quantitative analysis: the impact of ChatGPT on task engagement

Table 2 displays descriptive statistics for the learners' engagement in English argumentative essay writing for the pretest, immediate post-test, and delayed post-test. According to Table 2, initially, both the control group ($M = 83.85$, $SD = 14.17$) and the experimental group ($M = 83.50$, $SD = 12.19$) showed nearly the same level of engagement in writing English argumentative essays, as indicated by the closely related mean scores. After the intervention, the experimental group demonstrated a significant increase in their mean score ($M = 94.62$, $SD = 13.09$), indicating a substantial improvement in their engagement. While the control group also showed improvement ($M = 86.85$, $SD = 12.63$), their progress was not as significant as that of the experimental group. At the delayed post-test phase, the experimental group's mean score ($M = 93.03$, $SD = 11.40$) dropped slightly from the immediate post-test but remained high, indicating a sustained improvement in engagement over time. The control group, on the other hand, saw a minor increase ($M = 87.39$, $SD = 8.68$).

To determine if the observed differences were statistically significant, we conducted separate independent-samples *t*-tests. The results are presented in Table 3. As Table 3 shows, no significant difference was found between the control and experimental groups in their engagement before the intervention ($P = .91 > .05$). The effect size for the pretest was computed as 0.02, which is considered small and negligible (Plonsky and Oswald 2014). This confirms that the groups were statistically equivalent at the beginning of the study.

At the immediate post-test, the experimental group performed significantly better than the control group ($P = .023 < 0.05$). At the same time, its effect size was computed as 0.60, which falls into the medium range, according to Plonsky and Oswald (2014). This indicates a moderate effect of the intervention, with the experimental group showing a significantly larger increase in engagement, as compared to the control group immediately following the intervention.

At the delayed post-test, the two groups were statistically significant ($P = .03 < .05$). This suggests that the increase in the experimental group's engagement remained significant over time. The delayed post-test showed a medium effect size of 0.55 (Plonsky and Oswald 2014), suggesting that the intervention had a significant and lasting impact.

To examine whether the difference in each group's engagement was statistically significant across pre-treatment, post-treatment, and 1 month after the treatment, we conducted separate repeated-measures ANOVAs for both groups. The outcomes are presented in Tables 4–7. As shown

Table 2. Descriptive statistics.

	Group	N	Mean	Std. deviation	Std. Error mean
Pretest	Control	28	83.85	14.17	2.67
	Experimental	32	83.50	12.19	2.15
Immediate	Control	28	86.85	12.63	2.38
	Experimental	32	94.62	13.09	2.31
Delayed	Control	28	87.39	8.68	1.64
	Experimental	32	93.03	11.40	2.01

in Table 4, the repeated-measures ANOVA did not find a significant difference for the control group (F (1.91, 51.81) = 0.760, P = .468 > .05).

According to Table 5, the post hoc analysis results indicate that there was no significant difference in the control group's engagement across pre-treatment, post-treatment, and one month after the treatment (P = 1.000, .715 > .05), indicating that their level of engagement did not change notably at any point during the study.

Table 6 points to a significant difference for the experimental group (F (1.64, 50.98) = 8.794, P = .001 < .05).

According to Table 7, the post-hoc analysis showed a significant difference in the engagement of learners in the experimental group between pre-treatment and post-treatment (P = .001 < .05), as well as between pre-treatment and one month after the treatment (P = .001 < .05). However, the difference between post-treatment and one month after the treatment was not significant (P = 1.000 > .05). This indicates that their engagement changed significantly from the pretest to the immediate and delayed post-tests but not from the immediate post-test to the delayed one. In summary, the results suggest that using ChatGPT 4.0 had a significant and lasting positive impact on learners' engagement in writing English argumentative essays.

Qualitative analyses: how ChatGPT shapes task engagement

To determine how using ChatGPT 4.0 positively impacted learners' engagement in composing English argumentative essays, we ran a thematic analysis of their responses to the interview. Six

Table 3. Independent-samples T-tests

		Levene's test for equality of variances		*t*-test for equality of means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean difference	Std. error difference	95% confidence interval of the difference	
									Lower	Upper
Pretest	Equal variances assumed	.31	.57	.10	58	.917	.35	3.40	−6.45	7.17
	Equal variances not assumed			.10	53.69	.918	.35	3.43	−6.53	7.25
Immediate	Equal variances assumed	.11	.73	−2.33	58	.023	−7.76	3.33	−14.43	−1.09
	Equal variances not assumed			−2.33	57.42	.023	−7.76	3.32	−14.42	−1.11
Delayed	Equal variances assumed	2.63	.11	−2.13	58	.037	−5.63	2.64	−10.93	−0.34
	Equal variances not assumed			−2.16	56.97	.034	−5.63	2.59	−10.84	−0.43

key themes emerged from the analysis of the experimental group participants' responses. These include: 1) motivation, 2) reduced anxiety, 3) interest, 4) partnership, 5) personalized feedback, and 6) competence. Table 8 presents these themes along with their frequency and percentage.

As Table 8 shows, *motivation* emerged as the most prominent theme, corresponding to 26.83% of the participants. In this respect, Liam said:

1) *Well! The first effect it had was that it increased my motivation. Being alone, at first, I thought I could never write an actual essay because you know how difficult it is to write an essay. But later I got more motivated and tried harder and progressed.* (Liam, a 20-year-old second-year English learner)

As Excerpt 1 suggests, Liam mentioned that ChatGPT enhanced his motivation and willingness to persist when faced with challenges in essay writing. This is consistent with the SDT, which emphasizes the critical role of motivation, especially when originating from within the learner, in regulating behavior (Ryan and Deci 2017). According to SDT, learners like Liam are more likely to be engaged in educational settings, where they have control over their learning process

Table 4. Tests of within-subjects effects for the control group

Measure: Engagement in Essay Writing							
Source		**Type III sum of squares**	**df**	**Mean square**	**F**	**Sig.**	**Partial Eta squared**
Time	Sphericity Assumed	203.35	2	101.67	.76	.47	.02
	Greenhouse-Geisser	203.35	1.91	105.97	.76	.46	.02
	Huynh-Feldt	203.35	2.00	101.67	.76	.47	.02
	Lower-bound	203.35	1.00	203.35	.76	.39	.02
Error (Time)	Sphericity Assumed	7223.31	54	133.76			
	Greenhouse-Geisser	7223.31	51.81	139.41			
	Huynh-Feldt	7223.31	54.00	133.76			
	Lower-bound	7223.31	27.00	267.53			

Table 5. Pairwise comparisons for the control group

Measure: Engagement in Essay Writing						
(I) Time	**(J) Time**	**Mean difference (I-J)**	**Std. Error**	**Sig.**	**95% confidence interval for difference**	
					Lower bound	**Upper bound**
1	2	−3.00	3.39	1.00	−11.66	5.66
	3	−3.53	2.93	.71	−11.01	3.94
2	1	3.00	3.39	1.00	−5.66	11.66
	3	−0.53	2.92	1.00	−7.99	6.92
3	1	3.53	2.93	.71	−3.94	11.01
	2	.53	2.92	1.00	−6.92	7.99

(autonomy), face challenges that facilitate learning and progress (competence), and feel a sense of belonging to a learning community (relatedness).

The second most frequently mentioned theme was *reduced anxiety*, constituting 21.95% of the responses. Regarding reduced anxiety, Lillian pointed out:

Table 6. Tests of within-subjects effects for the experimental group

Measure: Engagement in Essay Writing							
Source		**Type III sum of squares**	**df**	**Mean square**	***F***	**Sig.**	**Partial Eta squared**
Time	Sphericity Assumed	2316.27	2	1158.13	8.79	.00	.22
	Greenhouse-Geisser	2316.27	1.64	1408.27	8.79	.00	.22
	Huynh-Feldt	2316.27	1.72	1342.90	8.79	.00	.22
	Lower-bound	2316.27	1.00	2316.27	8.79	.00	.22
Error (Time)	Sphericity Assumed	8165.06	62	131.69			
	Greenhouse-Geisser	8165.06	50.98	160.13			
	Huynh-Feldt	8165.06	53.46	152.70			
	Lower-bound	8165.06	31.00	263.38			

Table 7. Pairwise comparisons for the experimental group

Measure: Engagement in Essay Writing						
(I) Time	**(J) Time**	**Mean difference (I-J)**	**Std. Error**	**Sig.**	**95% confidence interval for difference**	
					Lower bound	**Upper bound**
1	2	−11.12	2.78	.00	−18.18	−4.06
	3	−9.53	2.27	.00	−15.28	−3.77
2	1	11.12	2.78	.00	4.06	18.18
	3	1.59	3.42	1.00	−7.08	10.26
3	1	9.53	2.27	.00	3.77	15.28
	2	.50	2.82	1.00	−6.45	7.47

Table 8. Thematic analysis of the interviews

Themes	**N**	**%**
Motivation	11	26.83
Reduced anxiety	9	21.95
Interest	7	17.07
Partnership	6	14.63
Personalized feedback	4	9.76
Competence	4	9.76
Total	41	100

2) *Well, I am a bit stressed myself. And part of this is because my English is not so good and I'm not really good at communicating. That's why I have a lot of stress in English classes. But I can confidently say that my stress is less because I always have access to someone who answers my questions and makes me better prepared and less stressed about language and writing.* (Lillian, a 23-year-old fourth-year English learner)

As the second excerpt indicates, Lillian's experience with ChatGPT led to a decline in her language learning anxiety. This is significant because language learning anxiety can negatively affect the learning process, as per Horwitz (2017). Thus, it is no surprise that decrease in her anxiety resulted in higher engagement levels. This finding is also in line with previous research that has linked technology-enhanced language learning with decreased anxiety levels (e.g. Bashori et al. 2021). Lillian's reflection suggests that ChatGPT served as a stress buffer, likely by offering the learner a sense of readiness and support. In their model of task engagement, Egbert *et al.* (2021) list support as one of the facilitators of engagement.

The next theme was *interest*, which was highlighted by 17.07% of the participants. In this respect, Emma said:

3) *The way I see it is that my interest in writing has increased because everything I need is right there, you know. Interest is important more than anything else for me. This tool made me spend more time and work harder. Of course, you may become too dependent on it which is a negative point. And of course, it has a negative effect on creativity. So, I guess what I'm saying is that we have to keep a balance so that we use both our own and ChatGPT's abilities.* (Emma, a 22-year-old third-year English learner)

Interest is a crucial factor for engagement, according to Egbert *et al.*'s (2021) task engagement model. Emma's experience highlights that while GenAI tools can spark interest and increase engagement, they can also hinder creativity if relied on too heavily. Her reflection echoes the concerns of scholars like Rudolph *et al.* (2023) and Zhang (2023), who emphasized the negative impact of excessive dependence on GenAI on academic, social, and career development. Therefore, as Emma pointed out, it is important to strike a balance between technology and traditional learning methodologies to enhance language learning without hindering the development of analytical skills.

The next theme was *partnership*, mentioned by 14.63% of the participants. In this respect, Natalie highlighted:

4) *Well, if we ignore the negative points like laziness and lack of creativity and things like that, I can say that for me it was more like a teammate. I mean, I didn't feel like I had to do such a difficult thing alone. Because it answered my questions and corrected my mistakes, I felt like I was writing with a fellow group member.* (Natalie, a 20-year-old third-year English learner)

The theme of partnership aligns with Vygotsky's sociocultural theory, specifically the concept of MKOs. According to Vygotsky (1978), learning is a social process, and cognitive development is largely influenced by social interactions. The MKO, according to Vygotsky, refers to someone who has a better understanding or a higher level of ability than the learner, particularly in a specific task, concept, or process. Traditionally the MKO is considered to be a teacher or an adult. Natalie's reflection suggests that ChatGPT acted as a peer-like MKO, offering assistance and support in learning. This partnership fostered an environment where the learner did not feel isolated and became more deeply engaged in the learning process.

Personalized feedback was another frequently mentioned theme, highlighted by 9.76% of the learners. Regarding this, Natalie said:

5) *What was very interesting to me was that it corrected my mistakes in writing. Well! One of our professors used to do that before, but it happened only once during the whole semester because there were so many of us, you know. But with AI, I can always have that experience and it's great because you learn so much.* (Mia, a 25-year-old fourth-year English learner)

Feedback, especially personalized feedback, is crucial for learning (Hattie and Timperley 2007; Olsen and Hunnes 2023) and Mia's experience indicates that this need is met by ChatGPT. This resonates with the current principles of learner-centered instruction in SLA. This form of instruction emphasizes the significance of adapting teaching methods to suit the individual needs, interests, and abilities of learners (Richards and Schmidt 2010). The approach is based on the realization that learners are unique and teaching should be adapted to suit their personal learning styles (Griffiths and Soruç 2021). Mia's statement underscores the advantage of GenAI in providing immediate and personalized feedback, which is not always feasible in traditional classroom settings, due to time constraints and teacher-to-learner ratios. Furthermore, the adaptability of ChatGPT provides the opportunity to give feedback on the subtleties of the target language that may not be apparent through more general comments.

Last, but not least, 9.76% of the learners who participated in the interviews identified *competence*. In this respect, David pointed out:

6) *Some of its ideas were not accurate. Yet, some of the suggestions were new and made me feel like I was learning something new. And one more thing! You know how we always say we should learn English in context and use English to learn it? It was like that. My conversations with ChatGPT improved some of my English skills such as speaking which was great. I must say, though, that its language is not very natural.* (David, a 22-year-old fourth-year English learner)

David and others' perception of *competence* suggests a sense of improved language skills through interaction with ChatGPT. This can be explained in terms of the SDT, which highlights the need for competence, along with autonomy and relatedness, as basic psychological needs. Meeting these needs leads to intrinsic motivation which in turn results in greater engagement.

Discussion

The discussion is organized into three main parts. First, we summarize the key findings. Second, we interpret these findings using established theoretical frameworks—including SDT, Vygotsky's sociocultural theory, and Egbert *et al.*'s (2021) task engagement model. Third, we compare and explore the potential reasons behind the observed patterns.

This investigation aimed to explore the effects of using ChatGPT 4.0 by L2 learners on their engagement in writing English argumentative essays. The results showed that working with ChatGPT significantly influenced learners' engagement in writing English argumentative essays across the study's timeline—from pre-intervention to immediate and delayed post-intervention. That is, ChatGPT is effective in enhancing learners' engagement in writing English argumentative essays in short and long terms. It does so by boosting their motivation, interest, and competence, decreasing their anxiety, creating a sense of partnership, and providing personalized feedback.

The decline in the learners' engagement at the delayed post-test might be due to their temporary overreliance on technological aid which later led to reduced independent effort from them. However, the richness of the technological interaction appears to offer substantial psychological and academic benefits in terms of motivation, support, sustained interest, and perceived academic partnership, as well as immediate feedback that facilitates learning and feelings of linguistic competence.

These findings invite deeper reflection on how GenAI tools like ChatGPT might reshape prevailing theoretical assumptions about engagement in second language writing contexts. Traditional frameworks often position engagement as a function of social interaction, structured instruction, or pedagogical design. In contrast, this study suggests that engagement can also emerge through personalized, autonomous interaction with AI, expanding the ways learners experience motivation and connection during writing. The learner's experience of ChatGPT as a "teammate" introduces an important dimension of human-machine collaboration in SLA, signaling a need to reconsider how relational and cognitive support are conceptualized in digital contexts.

The results can be explained in terms of SDT (Ryan and Deci 2017), Vygotsky's (1978) sociocultural theory, and Egbert *et al.*'s (2021) task engagement model. According to Self-Determination Theory (SDT), the ability to regulate one's own learning is essential and has a significant impact on learning behavior. This means that behaviors can range from the least determined to the most self-determined, depending on how self-directed they are. The most self-determined behaviors result in intrinsic motivation, where one completes a task for its own sake, which gives way to greater engagement (Mozgalina 2015). For intrinsic motivation to occur, autonomy, competence, and relatedness must be met as basic needs. In this study, autonomy was supported by the learners' perceived sense of control over their learning, as highlighted by Liam. Competence was supported by learners' perceived sense of achievement and learning as a result of the individualized feedback they received from ChatGPT, as noted by David and Mia. Finally, the partnership theme aligns with relatedness, creating a collaborative atmosphere where learners like Natalie don't feel alone in the learning process. This can also be interpreted in terms of Vygotsky's sociocultural theory, specifically the concept of MKO and ZPD. ChatGPT seems to serve as a technological MKO that assists learners in their ZPD. Natalie's feeling of working with a "teammate" indicates this kind of support. This challenges the traditional view of MKOs as exclusively human and suggests that certain AI tools may simulate the functions of human mediation by providing timely scaffolding, linguistic input, and dialogic interaction. Such reinterpretations of sociocultural learning support are particularly relevant in contemporary learning environments where digital mediation is becoming increasingly central.

Egbert *et al.*'s (2021) task engagement model is also a useful framework to interpret the findings. The model identifies six primary factors that can improve language task engagement when included in task design. These facilitators are authenticity, social interaction, learning support, interest, autonomy, and challenge. These facilitators were all identified by learners. Authenticity, broadly defined as learners' perceptions of the relevance and value of a task (Egbert *et al.* 2021), was noted by different learners, including Lillian, Emma, Natalie, and Mia. Additionally, Natalie's perception of ChatGPT as a teammate underscores the value of interaction in learning even if it is virtual. Furthermore, Lillian's experience with ChatGPT indicates how well it can serve as a learning support. As regards interest, ChatGPT's capability to engage learners with relevant material and feedback likely stimulated their interest and motivation, as Liam, Emma, and Mia noted. Concerning autonomy, as Liam highlighted, learners had control over their learning process, and interacting with ChatGPT only helped them in the face of challenges. Finally, regarding challenge, it may be said that although writing English essays is a highly challenging task, the customized feedback learners received from ChatGPT offered a balance between challenge and skill, as per Mia.

Overall, these findings suggest that GenAI tools have the potential to not only enhance task engagement but also expand its conceptual boundaries. Learner engagement is no longer solely mediated through teacher intervention or peer collaboration; instead, it can be sustained through real-time interaction with responsive, adaptive technologies. This shifts the landscape of task engagement in SLA, calling for a reexamination of how we define, support, and assess engagement in the age of AI-enhanced learning.

The results may lay the foundation for a more context-specific and refined framework of task engagement in technology-enhanced SLA settings, providing empirical data that could inform future theoretical developments. By integrating the quantitative evidence of enhanced engagement with qualitative data, the study highlights the potential of tools like ChatGPT to cater to the nuanced needs of learners when facing the complexities of English essay writing. The outcomes of this study may also inform the integration of GenAI tools, particularly ChatGPT, into L2 learning curricula by highlighting their potential to enhance learners' task engagement, even in the absence of structured guidance. Therefore, L2 educators may integrate ChatGPT 4.0 into L2 writing instruction to provide dynamic, personalized feedback that enhances learners' engagement. L2 teacher training programs may incorporate training on effectively using GenAI tools, ensuring that instructors can balance technological support with strategies that promote independent critical thinking. Additionally, L2 curriculum designers are encouraged to embed GenAI-driven activities into L2 writing curricula to create more engaging tasks that accommodate diverse learner needs. Finally, policymakers may develop ethical guidelines for adopting GenAI tools to prevent overreliance and support sustainable L2 learner-centered instruction.

Conclusion

The present investigation set out to explore if and how using ChatGPT 4.0 by L2 learners affects their engagement in writing English argumentative essays in the short and long terms. Overall, the findings suggest that ChatGPT can be an effective instructional tool for enhancing learners' engagement in writing English essays. It stimulates their motivation, interest, and competence while reducing their anxiety. Additionally, it fosters a sense of partnership and provides them with personalized feedback, making the writing process more engaging. The findings also highlight the need to maintain a balance between the advantages of using AI tools and the educational activities that encourage independent critical thinking and creativity.

Given the study's limitations, any conclusions must be tentative. The limitations include small and limited sample, reliance on self-reported measures, and the short span of data collection. Therefore, it is recommended that future studies recruit a large number of learners with diverse language proficiency levels, and cognitive, emotional, and social backgrounds. Future studies can also triangulate their data with other ways of measuring engagement, such as eye tracking (Dewan, Murshed and Lin 2019), as self-report measures only reveal learners' perceptions of their engagement without delving into their actual engagement. Furthermore, while the study collected data over a short span, longitudinal designs seem to be better suited to the dynamic nature of engagement. Finally, this study serves as an initial step toward understanding how GenAI influences engagement in L2 learning. Finding a balanced approach requires incorporating both technological and collaborative activities in L2 writing instruction. Hence, in future research, exploring AI as a supplementary tool rather than a replacement for peer interactions could yield richer insights into the dynamics of task engagement in technology-enhanced L2 writing instruction.

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Conflict of Interest

No potential conflict of interest was reported by the authors.

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